Useful Memory Has a Horizon

Temporal Validity and Structural Limits
in Finite-Memory Distribution Tracking under Drift

Vinicius Parreira

Independent Researcher

figueiredo@pm.me

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Abstract

Under drift, additional memory reduces finite-sample noise but also increases the age of the evidence used to estimate the present. Finite-memory distribution tracking is therefore governed by a temporal-validity horizon. A general horizon law separates finite-memory tracking error into a finite-sample term, determined by the chosen geometry and estimator, and a drift-induced staleness term, determined by the regularity of the target path:

$$\mathbb{E}\, d\left(\widehat{P}_t^{(n)}, P_t\right) \le C_K n^{-a} + C_S \zeta n^H.$$

Here $a > 0$ is the finite-sample rate exponent and $H \in (0, 1]$ is the temporal roughness exponent. Optimizing this balance yields the temporal-validity horizon $n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}$.

A tractable one-dimensional bounded-support fixed-span triangular-array model provides a rigorous root-n finite-sample regime through quantile/Bahadur structure under an explicit integrated remainder hypothesis. A Gaussian lower-bound construction shows that the horizon is structural rather than an artifact of a particular estimator, and the same object admits a Gaussian location minimax benchmark on deterministic Hölder paths. For fixed-ε Sinkhorn geometry, a pointwise one-dimensional fixed-support inheritance result is available; on compact ε-bands, the same proof-model argument yields a uniform one-dimensional band law, while the broader embedded fixed-span horizon inheritance on nontrivial regularization bands remains an explicit conjecture.

Memory under drift is governed by temporal validity, not by a universal window rule. The horizon depends jointly on finite-sample geometry and temporal roughness, and retained evidence can become invalid for the present before change becomes statistically detectable.

Keywords: tracking under drift; temporal validity; finite-memory estimation; distribution tracking; Wasserstein distance; Sinkhorn divergence.
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1 Introduction

In stationary statistics, memory is an unambiguous resource: larger samples reduce variance because the target distribution does not change. Under drift, that logic fails. In streaming estimation, monitoring, and online adaptation, the evidence retained in memory was generated by distributions that may no longer represent the present. More memory therefore does two things at once: it reduces finite-sample noise, and it increases the age of the evidence used to estimate the current state.

That tension creates a horizon problem. Short windows are fresh but noisy. Long windows are stable but stale. The temporal-validity horizon asks how long retained evidence remains useful for estimating the present before drift-induced misalignment overtakes the remaining variance gain.

A general horizon law captures this problem. The starting point is an abstract decomposition

$$\mathbb{E}\, d\left(\widehat{P}_t^{(n)}, P_t\right) \le C_K n^{-a} + C_S \zeta n^H,$$

where $a > 0$ is the finite-sample rate exponent of the chosen geometry and estimator, $H \in (0, 1]$ is the temporal roughness exponent of the drift path, and ζ is the roughness amplitude. Optimizing this balance yields the temporal-validity horizon

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

The horizon depends jointly on finite-sample geometry and on the temporal regularity with which drift accumulates staleness.

The paper develops this object in four steps. First, it establishes the law in a tractable one-dimensional bounded-support fixed-span triangular-array model. In that regime, the quantile representation of W_2^2, a triangular-array Bahadur expansion, and a design-effect comparison yield the root-n finite-sample rate

$$\mathbb{E}\, W_2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = O(n^{-1/2}).$$

under explicit compact-support, fixed-span, and integrated-remainder hypotheses. This gives a rigorous first proof model for the general horizon law.

Second, it proves a structural lower bound showing that the horizon is imposed by the drift geometry itself rather than by one particular estimator. A refined Gaussian lower-bound construction sharpens the constant-level geometry, and in the Gaussian location model the same object yields a full minimax benchmark on deterministic Hölder paths.

Third, it isolates the proved structural ingredients for fixed-ε Sinkhorn geometry on an embedded fixed-span model, proves pointwise and bandwise one-dimensional fixed-ε inheritance results, and states the broader embedded triangular-array band inheritance as an explicit conjecture.

Fourth, it formulates the broader family-level inheritance question: which regular distribution families and operational geometries support which finite-sample rates, and therefore which members of the horizon family. Memory under drift is governed by the interaction between finite-sample geometry and temporal roughness, not by a single universal window prescription.

The general horizon law, the uniform-window staleness control, the one-dimensional proof-model upper bound, the Gaussian structural lower bound, and the Gaussian location benchmark are theorem-level results. The fixed-ε Sinkhorn section proves structural ingredients together with pointwise and bandwise one-dimensional fixed-ε inheritance results, and states the broader embedded triangular-array band law as a conjecture. The numerical evidence concentrates on moderate compact bands bounded away from zero. Broader family-level inheritance remains open.

The empirical section makes the finite-memory U-curve visible, shows roughness-dependent horizon scaling, reports evidence relevant to the fixed-ε Sinkhorn conjecture, and exhibits a measurable lag between validity loss and the first detector alarm. Temporal validity and changepoint evidence are different statistical objects: retained evidence can stop representing the present before change becomes detectable. The experiments are synthetic and structurally targeted; their role is to reveal the predicted signatures, not to settle a benchmark comparison against adaptive tracking baselines.

This framing is adjacent to several literatures—dynamic regret, adaptive windowing, locally stationary smoothing, adaptive filtering, and distributional tracking under transport geometry—but the question here is different. Dynamic-regret analysis asks how well one tracks a moving comparator, and adaptive-window or drift-detection methods ask when the accumulated evidence is sufficient to react. Here the question is: for finite-memory tracking under drift, how long does retained evidence remain temporally valid for estimating the present distribution before age becomes the dominant source of error?

Section 2 develops the general horizon law, drift-regularity bounds, a tractable proof model, a structural lower bound, and the benchmark and operational regimes. Section 3 presents the corresponding empirical signatures. Section 4 discusses related work, and Section 5 discusses scope, limitations, and open problems.

2 A Horizon Law for Finite-Memory Tracking under Drift

This section fixes the finite-memory objects and develops the horizon law, structural lower bound, benchmark theorem, and conjectural extensions.

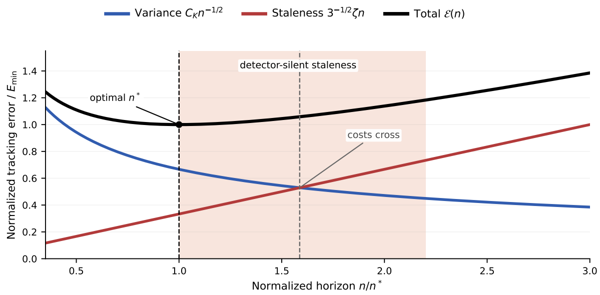

Figure 1: Two clocks of drift. The finite-sample term falls with horizon, the staleness term rises with age, and total error is minimized at the temporal-validity horizon n^*. The shaded region marks pre-detection validity loss: retained evidence can stop representing the present before enough changepoint evidence accumulates to trigger an alarm.

For a finite-memory estimator with lag weights $w_0, \ldots, w_{n-1}$, the natural age variable is the effective lag

$$A(w) := \sum_{j=0}^{n-1} j\, w_j .$$

Its corresponding staleness at time t is the weighted transport misalignment

$$S_t(w) := \sum_{j=0}^{n-1} w_j\, W_2(P_{t-j}, P_t)\, .$$

For a uniform window, $A(w) = (n-1)/2$, so window length and temporal age are the same object up to constants.

Throughout, P_t denotes the present target law, $\bar{P}_t^{(n)}$ the window target obtained by averaging the last n target laws, and $\widehat{P}_t^{(n)}$ the estimator built from the last n observations. The finite-sample term always refers to the estimation error relative to $\bar{P}_t^{(n)}$, while the staleness term refers to the discrepancy between $\bar{P}_t^{(n)}$ and P_t.

The theory combines three ingredients: a finite-sample bound relative to $\bar{P}_t^{(n)}$, a path-class control on the staleness between $\bar{P}_t^{(n)}$ and P_t, and, when available, a lower-bound or benchmark theorem showing that the resulting horizon is structural on the geometry under study.

2.1 General Horizon Law

The main theorem separates finite-sample behavior from drift-induced staleness.

Theorem 2.1 (Abstract upper law)**.** *Let d be a probability metric satisfying the triangle inequality. At time t, let P_t be the present target law, let $\bar{P}_t^{(n)}$ be the window target, and let $\widehat{P}_t^{(n)}$ be the estimator built from the last n observations. Assume that for some constants $C_K > 0$, $C_S > 0$, exponent $a > 0$, roughness index $H \in (0, 1]$, and amplitude $\zeta > 0$,*

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) \le C_K n^{-a}$$

and

$$d\Big(\bar{P}_t^{(n)}, P_t\Big) \le C_S \zeta n^H$$

hold for all n in the regime of interest. Then

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_K n^{-a} + C_S \zeta n^H.$$

Proof. By the triangle inequality,

$$d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le d\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) + d\Big(\bar{P}_t^{(n)}, P_t\Big).$$

Taking expectations and applying the two assumptions yields the claim. □

The proof is immediate, but the theorem isolates the structure of the problem: the path class supplies the staleness term, while the chosen geometry and sample regime supply the finite-sample term.

Corollary 2.2 (Optimized temporal-validity horizon)**.** *Under the assumptions of theorem 2.1, the upper envelope*

$$\Phi(n) := C_K n^{-a} + C_S \zeta n^H$$

is minimized, up to constant factors depending only on a, H, and C_S, at

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)},$$

and the corresponding optimized error scale is

$$\Phi(n^*) \asymp C_K^{H/(a+H)} \zeta^{a/(a+H)}.$$

Proof. Balance the decreasing finite-sample term against the increasing staleness term:

$$C_K n^{-a} \asymp C_S \zeta n^H.$$

This gives $n^{a+H} \asymp C_K/\zeta$, hence the displayed horizon law. Substituting that scale back into either term gives the optimized error scale. □

The horizon depends jointly on the finite-sample rate exponent and the roughness exponent. Different geometries and different drift regularities induce different memory scales.

2.2 Drift Regularity and Staleness Accumulation

The staleness side of the law is controlled by a deterministic path class. For $H \in (0, 1]$ and $\zeta > 0$, write

$$\mathfrak{P}_H(\zeta) := \left\{(P_s)_{s\in\mathbb{Z}} : W_2(P_{t-j}, P_t) \le \zeta j^H \text{ for all } j \ge 0\right\}.$$

The case $H = 1$ is the local Lipschitz envelope; smaller H correspond to slower accumulation of staleness with lag.

Lemma 2.3 (Hölder staleness bound via averaged couplings)**.** *Let*

$$\bar{P}_t^{(n)} := \frac{1}{n} \sum_{j=0}^{n-1} P_{t-j}.$$

If the target path belongs to $\mathfrak{P}_H(\zeta)$, then

$$W_2^2\Big(\bar{P}_t^{(n)}, P_t\Big) \le \frac{\zeta^2}{n} \sum_{j=0}^{n-1} j^{2H},$$

and therefore

$$W_2\Big(\bar{P}_t^{(n)}, P_t\Big) \le \zeta \left(\frac{1}{n} \sum_{j=0}^{n-1} j^{2H}\right)^{1/2} = C_{H,n}\, \zeta n^H,$$

where

$$C_{H,n} := \left(n^{-(2H+1)} \sum_{j=0}^{n-1} j^{2H}\right)^{1/2}.$$

In particular, $C_{H,n} \to (2H+1)^{-1/2}$ as $n \to \infty$.

Proof. For each lag j, let π_j be an optimal coupling between P_{t-j} and P_t. Their average

$$\bar{\pi} := \frac{1}{n} \sum_{j=0}^{n-1} \pi_j$$

couples $\bar{P}_t^{(n)}$ with P_t, so

$$\begin{aligned} W_2^2\Big(\bar{P}_t^{(n)}, P_t\Big) &\le \int \|x - y\|^2 \, d\bar{\pi}(x, y) \\ &= \frac{1}{n} \sum_{j=0}^{n-1} W_2^2(P_{t-j}, P_t). \end{aligned}$$

Applying the Hölder envelope gives the displayed finite-n constant, and taking square roots completes the proof. □

Combining lemma 2.3 with corollary 2.2 yields the horizon family

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

In the root-n finite-sample regime $a = 1/2$,

$$n^*(1/2, H) \asymp (C_K/\zeta)^{2/(1+2H)}.$$

The finite-n constant in lemma 2.3 is the constant used here for uniform windows; the simpler $(2H+1)^{-1/2}$ expression is its large-window limit.

2.3 A Tractable One-Dimensional Proof Model

The abstract law needs at least one rigorous finite-sample instantiation. The bounded-support fixed-span one-dimensional triangular array provides a tractable proof model for that role. Its scope is intentionally narrow: common compact support, uniformly regular window mixtures, and an explicit integrated Bahadur remainder hypothesis.

Proposition 2.4 (Root-n finite-sample bound in the 1D proof model)**.** *Let $X_{n,1}, \dots, X_{n,n}$ be independent one-dimensional observations with laws $P_{n,1}, \dots, P_{n,n}$, and define the window mixture*

$$\bar{P}_n = \frac{1}{n} \sum_{j=1}^{n} P_{n,j}.$$

Assume:

(i) all $P_{n,j}$ are supported on a common compact interval $[L, U]$;

(ii) the within-window drift span is uniformly bounded in n;

(iii) each mixture law $\bar{P}_n$ has a density $\bar{f}_n$ on $[L, U]$ that is bounded below by a positive constant and uniformly Hölder;

(iv) writing q_n for the quantile function of $\bar{P}_n$ and $\widehat{q}_n$ for the empirical quantile function of $\widehat{P}_n^{\mathrm{tri}}$, one has the Bahadur representation

$$\widehat{q}_n(u) - q_n(u) = \frac{u - \widehat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u), \qquad u \in (0,1),$$

with

$$\mathbb{E}\int_0^1 R_n(u)^2\,du = o(n^{-1}).$$

Then

$$\mathbb{E}\,W_2^2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = O(n^{-1}),$$

and therefore

$$\mathbb{E}\,W_2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = O(n^{-1/2}).$$

Proof. In one dimension,

$$W_2^2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = \int_0^1 \left(\widehat{q}_n(u) - q_n(u)\right)^2 du,$$

so the problem reduces to the integrated L_2 error of the empirical quantile process. By item (iv),

$$\widehat{q}_n(u) - q_n(u) = \frac{u - \widehat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u).$$

Define

$$Z_n(u) := \frac{u - \widehat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))}.$$

Then

$$\mathbb{E}\int_0^1 \left(\widehat{q}_n(u) - q_n(u)\right)^2 du = \mathbb{E}\int_0^1 \left(Z_n(u) + R_n(u)\right)^2 du.$$

Let $m > 0$ be a uniform lower bound for $\bar{f}_n$. Since

$$\widehat{F}_n(x) = \frac{1}{n}\sum_{j=1}^{n} \mathbf{1}\{X_{n,j} \le x\},$$

independence across the row gives

$$\mathrm{Var}(\widehat{F}_n(q_n(u))) = \frac{1}{n^2}\sum_{j=1}^{n} p_{n,j}(u)(1 - p_{n,j}(u)) \le \frac{1}{4n},$$

where $p_{n,j}(u) = P_{n,j}((-\infty, q_n(u)])$. Therefore

$$\mathbb{E}\int_0^1 Z_n(u)^2 du \le m^{-2}\int_0^1 \mathrm{Var}(\widehat{F}_n(q_n(u)))du \le \frac{1}{4m^2 n}.$$

By item (iv),

$$\mathbb{E}\int_0^1 R_n(u)^2 du = o(n^{-1}).$$

The cross term is then $o(n^{-1})$ by Cauchy–Schwarz, since $\mathbb{E}\int_0^1 Z_n(u)^2 du = O(n^{-1})$ and $\mathbb{E}\int_0^1 R_n(u)^2 du = o(n^{-1})$. Hence the expected squared Wasserstein error is $O(n^{-1})$, and Jensen's inequality gives the root-n rate for $\mathbb{E}\,W_2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right)$.

The main conclusion is the exponent. Under this bounded-support fixed-span regime, the triangular-array finite-sample term remains root-n. ☐

Remark 2.5 (Sufficient conditions for the proof model). The only nontrivial assumption in proposition 2.4 is item (iv). For i.i.d. quantile processes, Bahadur representations of this type are classical; see Bahadur [1966] and van der Vaart [1998, Chapter 21]. The present paper uses the triangular-array version as an explicit hypothesis for the fixed-span proof model rather than claiming a general theorem that verifies it. This covers, for example, bounded-support smooth location perturbations of a common base density and bounded smooth monotone transforms of such a base law, provided the fixed-span condition is maintained and the stated integrated remainder bound holds. A broad verification theorem for independent non-identically distributed rows is left open. Separate diagnostics on a smooth fixed-support deformation model show integrated residual decay faster than n^{-1} across the tested span and roughness values, which supports the plausibility of item (iv) on that class but is not used as proof.

2.4 Structural Lower Bound

The upper law would be substantially weaker if it were only the optimum of one estimator. The lower bound shows that, in the root-n finite-sample regime $a = 1/2$, the horizon is imposed by the drift geometry itself.

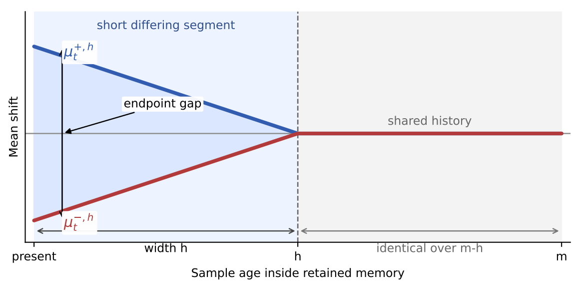

Figure 2: Gaussian lower-bound construction. The two paths share most of the retained history and differ only on a recent segment of width h, yet remain separated at the present. Optimizing that recent segment already produces the same horizon exponent as the upper law in the root-n regime.

Proposition 2.6 (Structural lower bound in the root-n regime)**.** *Fix $H \in (0,1]$ and work on the Gaussian location subclass with known variance σ^2 and path pair*

$$\mu_{t-j}^{\pm,h} = \pm\beta(h-j)_+^H, \qquad j = 0, 1, \ldots, h,$$

where $\beta \le \zeta$ enforces the roughness budget. Let $\mathcal{G}_{H,\zeta}^{\mathrm{Gauss}}$ denote the corresponding Gaussian location subclass of target paths, and suppose the retained observations are independent with laws $N(\mu_{t-j}^{\pm,h}, \sigma^2)$. Then there exists a *constant $c_H > 0$ depending only on H such that the endpoint tracking risk over this subclass satisfies*

$$\inf_{\widehat{P}} \sup_{P \in \mathcal{G}^{\mathrm{Gauss}}_{H,\zeta}} \mathbb{E}\, W_2\left(\widehat{P}_t, P_t\right) \geq c_H\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Consequently, in the root-n finite-sample regime $a = 1/2$, the horizon scale

$$h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$$

is structural even on this restricted subclass.

Proof outline. Compare two hypotheses generated by the path pair $\mu^{+,h}_{t-j}$ and $\mu^{-,h}_{t-j}$. At the present time, the endpoint separation is of order βh^H. On the Gaussian location subclass, this is also the W_2 separation. The Kullback–Leibler divergence between the two hypotheses is proportional to

$$\beta^2 \sum_{r=1}^{h} r^{2H}/\sigma^2.$$

Choosing

$$\beta = \min\left\{\zeta, \frac{\sigma}{2\sqrt{\sum_{r=1}^{h} r^{2H}}}\right\}$$

keeps the testing problem in the Le Cam regime while respecting the roughness budget. The resulting two-point lower bound is of order βh^H. Balancing the active roughness constraint against the testing constraint gives $h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$ and therefore the displayed lower law. The exact Gaussian two-point calculation used here is recorded in appendix A.

The lower bound is subclass based rather than class-tight, but that is enough for the central claim: the horizon is a structural feature of the problem rather than an estimator artifact.

The proposition above is the lower-bound fact needed for the main horizon law. The same Gaussian construction also exposes a sharper constant-level geometry. The next proposition records the best constant-level refinement presently available within that proof scheme.

2.5 Refined Gaussian Lower-Bound Geometry

Let Φ and φ denote the standard normal distribution function and density.

Proposition 2.7 (Refined Gaussian lower-bound constants). *Fix $H \in (0, 1]$ and set $p_H \coloneqq 2H/(2H + 1)$. Let $x_H > 0$ be the unique maximizer of $x^{p_H}\Phi(-x)$, equivalently the unique solution of*

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

For the Gaussian ramp profile $\mu^{\pm,h}_{t-j} = \pm\zeta(h - j)^H_+$, let

$$L^{\mathrm{ramp}}_h(\sigma, \zeta) \coloneqq \zeta h^H \Phi\left(-\frac{\zeta}{\sigma}\sqrt{\sum_{r=1}^{h} r^{2H}}\right).$$

Then, as $\sigma/\zeta \to \infty$,

$$\sup_{h\geq 1} L^{\mathrm{ramp}}_h(\sigma, \zeta) \sim C^{\mathrm{ramp}}_H\, \sigma^{2H/(2H+1)}\zeta^{1/(2H+1)},$$

with

$$C^{\mathrm{ramp}}_H = (2H + 1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

Among all endpoint-saturating Hölder profiles, the profile

$$g^{\min}_r = h^H - (h - r)^H$$

minimizes the testing energy and has asymptotic energy constant

$$I_H = \int_0^1 \left(1 - (1 - u)^H\right)^2 du = \frac{2H^2}{(H + 1)(2H + 1)}.$$

If

$$L^{\min}_h(\sigma, \zeta) \coloneqq \zeta h^H \Phi\left(-\frac{\zeta}{\sigma}\sqrt{\sum_{r=1}^{h} (g^{\min}_r)^2}\right),$$

then, as $\sigma/\zeta \to \infty$,

$$\sup_{h\geq 1} L^{\min}_h(\sigma, \zeta) \sim C^{\min}_H\, \sigma^{2H/(2H+1)}\zeta^{1/(2H+1)},$$

with

$$C^{\min}_H = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

For $H < 1$, this strictly improves the ramp constant by the factor

$$\frac{C^{\min}_H}{C^{\mathrm{ramp}}_H} = \left(\frac{H + 1}{2H^2}\right)^{H/(2H+1)} > 1,$$

while at $H = 1$ the two profiles coincide.

Proof. The exact Gaussian testing problem reduces the two-point lower bound to the displayed quantities L^{ramp}_h and $L^{\min}_h$. After rescaling by $h = A(\sigma/\zeta)^{2/(2H+1)}$, the ramp energy and endpoint-minimal energy have limits $(2H + 1)^{-1}$ and I_H, respectively. The corresponding normalized payoff becomes a scalar multiple of $x^{p_H}\Phi(-x)$, whose unique maximizer is x_H. The endpoint-minimal profile minimizes testing energy among all endpoint-saturating Hölder profiles, so it yields the sharper constant. Full details are given in appendix A. □

These refinements sharpen the constant-level lower-bound picture, but the main theoretical role of this section is narrower: they strengthen the Gaussian construction without changing the horizon exponents or the status of the main results.

The same refined lower-bound geometry also yields a full class-level benchmark once the model is specialized to Gaussian location tracking. In that setting, the deterministic Hölder envelope, the finite-memory estimator, and the lower-bound construction all live in the same one-dimensional parameterization, so this model admits a direct minimax rate statement.

Theorem 2.8 (Gaussian location minimax benchmark on the deterministic Hölder class). *Fix $H \in (0, 1]$, $\zeta > 0$,* *and $\sigma > 0$. Let $\mathfrak{M}_H(\zeta)$ denote the class of deterministic mean paths $(\mu_s)_{s\in\mathbb{Z}}$ such that*

$$|\mu_{t-j} - \mu_t| \le \zeta j^H \qquad \text{for all } j \ge 0.$$

Suppose the observations are independent with

$$Y_{t-j} \sim N(\mu_{t-j}, \sigma^2), \qquad j = 0, 1, \ldots, n-1,$$

and let the target law at time t be $P_t = N(\mu_t, \sigma^2)$. Define the optimized finite-memory minimax risk

$$\mathcal{R}_H^*(\sigma, \zeta) := \inf_{n\ge 1} \inf_{\widehat{P}_t^{(n)}} \sup_{\mu\in\mathfrak{M}_H(\zeta)} \mathbb{E}\, W_2\Big(\widehat{P}_t^{(n)}, P_t\Big).$$

Then there exist constants $0 < c_H \le C_H < \infty$, depending only on H, such that

$$c_H\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)} \le \mathcal{R}_H^*(\sigma, \zeta) \le C_H\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

In particular, the deterministic Hölder class has horizon scale

$$n_H^* \asymp (\sigma/\zeta)^{2/(2H+1)}.$$

Moreover, the uniform-window mean estimator has an exact asymptotic upper constant on this benchmark. If $y_H > 0$ solves

$$H y_H \big(2\Phi(y_H) - 1\big) = \varphi(y_H),$$

then its asymptotic horizon shape is

$$A_H^{\mathrm{mean}} = ((H+1)y_H)^{2/(2H+1)}$$

and its asymptotic risk constant is

$$U_H^{\mathrm{mean}} = ((H+1)y_H)^{-1/(2H+1)} \times \Big(2\varphi(y_H) + y_H\big(2\Phi(y_H) - 1\big)\Big).$$

Proof. For the upper bound, estimate the present mean by the uniform-window average

$$\widehat{\mu}_t^{(n)} := \frac{1}{n}\sum_{j=0}^{n-1} Y_{t-j}$$

and set $\widehat{P}_t^{(n)} = N(\widehat{\mu}_t^{(n)}, \sigma^2)$. Because equal-variance Gaussian laws satisfy

$$W_2\big(N(m_1, \sigma^2), N(m_2, \sigma^2)\big) = |m_1 - m_2|,$$

the risk reduces to mean estimation error. The noise term is $\sigma\sqrt{2/(\pi n)}$, while the deterministic bias is bounded by

$$\frac{\zeta}{n}\sum_{j=0}^{n-1} j^H,$$

so

$$\sup_{\mu\in\mathfrak{M}_H(\zeta)} \mathbb{E}\, W_2\Big(\widehat{P}_t^{(n)}, P_t\Big) \le \sigma\sqrt{\frac{2}{\pi n}} + \frac{\zeta}{n}\sum_{j=0}^{n-1} j^H.$$

Rescaling by $n = A(\sigma/\zeta)^{2/(2H+1)}$ yields a one-parameter normalized upper envelope. Differentiating its exact Gaussian-location form gives the stated root equation for y_H, and therefore the displayed asymptotic upper constant and horizon shape. In particular, this proves the stated upper rate and horizon scale.

For the lower bound, use the symmetric two-point subfamily

$$\mu_{t-j}^{\pm,h} = \pm\beta\big(h^H - j^H\big)_+, \qquad j \ge 0,$$

with $\beta \le \zeta$. This family lies in $\mathfrak{M}_H(\zeta)$ because

$$|\mu_t^{\pm,h} - \mu_{t-j}^{\pm,h}| = \beta\min(h^H, j^H) \le \zeta j^H.$$

Moreover, for fixed endpoint value βh^H, it minimizes the Gaussian testing energy over all symmetric lag profiles allowed by the present-point Hölder envelope, since each lag coordinate is independently minimized by $\beta(h^H - j^H)_+$. The exact Gaussian two-point calculation from the refined lower-bound analysis in appendix A therefore applies on the full deterministic Hölder class and yields a lower bound of order

$$\sigma^{2H/(2H+1)}\zeta^{1/(2H+1)}.$$

Optimizing over h gives the same horizon scale $(\sigma/\zeta)^{2/(2H+1)}$. Combining the two sides proves the theorem. □

2.6 Benchmark and Operational Regimes

The one-dimensional proof model is the first rigorous finite-sample instantiation. Two broader settings matter for scope: a Gaussian benchmark on a full path class, and an operational Sinkhorn regime in which structural ingredients are proved but the full inheritance theorem is conjectural. The linear-Gaussian entropic-OT setting is narrower still: its Kalman/Riccati structure suggests a constructive intermediate operational regime with an explicit recursion, sitting between the i.i.d. fixed-ε theory and the embedded triangular-array conjecture [Kalman, 1960, Jazwinski, 1970, Akyildiz et al., 2024].

Operational regime. For fixed-ε debiased Sinkhorn, the question is whether a practical geometry can supply a stable finite-sample term on the embedded fixed-span model. The intended theorem shape is

$$\mathbb{E}\, S_\varepsilon\Big(\widehat{P}_{t,\mathrm{tri}}^{(n)}, \bar{P}_t^{(n)}\Big) \le C_\varepsilon n^{-a_\varepsilon}$$

under bounded support, fixed span, and structural low-dimensionality, where S_ε denotes a fixed-ε Sinkhorn-type geometry and the exponent a_ε matches the corresponding i.i.d. mixture benchmark. The structural part of this setup is proved, but the full triangular-array inheritance statement is not.

Definition 2.9 (Fixed-ε operational setting)**.** Fix a bounded embedded fixed-span model, an intrinsic complexity index k, a dual Hölder smoothness level α, and an operational band $\mathcal{B} \subset (0, \infty)$ of regularization values. We say that this model is in the fixed-ε operational setting if:

(i) the i.i.d. fixed-ε benchmark on the same support class admits a finite-sample bound with exponent a_ε on $\mathcal{B}$;

(ii) the triangular-window and i.i.d.-mixture supports have identical operational covering complexity on $\mathcal{B}$;

(iii) the dual class lies in the parametric adaptation region $\alpha > k/2$.

Proposition 2.10 (Structural ingredients for fixed-ε Sinkhorn)**.** *On the bounded embedded fixed-span model with squared-Euclidean cost, the first three items in definition 2.9 are structural: the support-complexity inheritance is exact, and the dual-complexity condition is precisely the LCA threshold $\alpha > k/2$.*

Proposition 2.11 (Pointwise fixed-ε inheritance in the 1D proof model)**.** *Assume the hypotheses of proposition 2.4, let $\widehat{P}_n^{\mathrm{iid}}$ be the empirical measure of n i.i.d. samples from $\bar{P}_n$, and fix $\varepsilon > 0$. For the debiased Sinkhorn divergence S_ε with squared-Euclidean cost on the common compact support, one has*

$$\mathbb{E}\, S_\varepsilon\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = O(n^{-1/2}).$$

Proof. Write $V_\varepsilon(\mu,\nu)$ for the regularized OT value so that

$$S_\varepsilon(\mu,\nu) = V_\varepsilon(\mu,\nu) - \tfrac{1}{2}V_\varepsilon(\mu,\mu) - \tfrac{1}{2}V_\varepsilon(\nu,\nu).$$

Because all measures are supported on one compact interval, they lie in a common bounded quadratic-moment class. By Eckstein and Nutz [2022, Thm. 3.7(ii), Ex. 3.4], for fixed ε there exists $L_\varepsilon < \infty$ depending only on that class and on the cost such that

$$|V_\varepsilon(\mu,\nu) - V_\varepsilon(\mu',\nu')| \le L_\varepsilon\big(W_2(\mu,\mu') + W_2(\nu,\nu')\big).$$

Applying this bound to the cross term and both self terms gives

$$|S_\varepsilon(\mu,\nu) - S_\varepsilon(\mu',\nu')| \le 2L_\varepsilon\big(W_2(\mu,\mu') + W_2(\nu,\nu')\big).$$

With $\nu = \nu' = \bar{P}_n$,

$$S_\varepsilon\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) \le S_\varepsilon\left(\widehat{P}_n^{\mathrm{iid}}, \bar{P}_n\right) + 2L_\varepsilon W_2\left(\widehat{P}_n^{\mathrm{tri}}, \widehat{P}_n^{\mathrm{iid}}\right).$$

Taking expectations and using the triangle inequality gives

$$\mathbb{E}\, W_2\left(\widehat{P}_n^{\mathrm{tri}}, \widehat{P}_n^{\mathrm{iid}}\right) \le \mathbb{E}\, W_2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) + \mathbb{E}\, W_2\left(\widehat{P}_n^{\mathrm{iid}}, \bar{P}_n\right).$$

The first term is $O(n^{-1/2})$ by proposition 2.4. The second is the classical i.i.d. one-dimensional quantile-process rate on the same compact support class [Bahadur, 1966, van der Vaart, 1998], so it is also $O(n^{-1/2})$. Hence

$$\mathbb{E}\, W_2\left(\widehat{P}_n^{\mathrm{tri}}, \widehat{P}_n^{\mathrm{iid}}\right) = O(n^{-1/2}).$$

Finally, under the null comparison with fixed ε and compact support, the i.i.d. debiased Sinkhorn term satisfies

$$\mathbb{E}\, S_\varepsilon\left(\widehat{P}_n^{\mathrm{iid}}, \bar{P}_n\right) = O(n^{-1})$$

by del Barrio et al. [2023, Thm. 5.1, after fixed-ε rescaling]; see also the null-limit law in Goldfeld et al. [2024]. Combining the bounds yields the displayed $O(n^{-1/2})$ rate. □

Proposition 2.12 (Bandwise fixed-ε inheritance in the 1D proof model)**.** *Let $\mathcal{B} = [\varepsilon_{\min}, \varepsilon_{\max}] \subset (0,\infty)$ be a compact regularization band. Under the hypotheses of proposition 2.11 on a common compact support class, there exists $C_{\mathcal{B}} < \infty$ such that*

$$\sup_{\varepsilon\in\mathcal{B}} \mathbb{E}\, S_\varepsilon\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) \le C_{\mathcal{B}} n^{-1/2}.$$

Consequently, the induced horizon scale is uniform on $\mathcal{B}$:

$$\sup_{\varepsilon\in\mathcal{B}} n_\varepsilon^*(H) \asymp (C_{\mathcal{B}}/\zeta)^{1/(H+1/2)}.$$

Proof. Repeat the proof of proposition 2.11. The stability constant L_ε from Eckstein and Nutz [2022, Thm. 3.7(ii), Ex. 3.4] is bounded on $\mathcal{B}$ because all measures live in one compact support class and $\varepsilon \ge \varepsilon_{\min} > 0$. The i.i.d. debiased-Sinkhorn benchmark on compact support is also uniform on $\mathcal{B}$ by the fixed-ε asymptotics of Goldfeld et al. [2024] and the compact-support null bounds of del Barrio et al. [2023, Thm. 5.1, after fixed-ε rescaling]. Since the triangular-to-i.i.d. W_2 comparison is independent of ε, the constants in the pointwise proof can be chosen uniformly over $\mathcal{B}$. □

Conjecture 2.13 (Bandwise fixed-ε Sinkhorn horizon inheritance)**.** *Beyond the one-dimensional fixed-ε band result above, assume the fixed-ε operational setting from definition 2.9. Then, on bands where the i.i.d. benchmark stays in the parametric Sinkhorn regime, the triangular-array finite-sample term should satisfy*

$$\mathbb{E}\, S_\varepsilon\left(\widehat{P}_{t,\mathrm{tri}}^{(n)}, \bar{P}_t^{(n)}\right) \le C_\varepsilon n^{-1/2}, \qquad \varepsilon \in \mathcal{B},$$

and the corresponding horizon scale should satisfy

$$n_\varepsilon^*(H) \asymp (C_\varepsilon/\zeta)^{1/(H+1/2)}.$$

More generally, if the i.i.d. benchmark on $\mathcal{B}$ has exponent a_ε, then the operational horizon law should be

$$n_\varepsilon^*(a_\varepsilon, H) \asymp (C_\varepsilon/\zeta)^{1/(a_\varepsilon+H)}.$$

Evidence. The i.i.d. benchmark comes from existing fixed-ε entropic transport results, including fixed-marginal limit theory for the Sinkhorn divergence and intrinsic-complexity adaptation [Goldfeld et al., 2024, Stromme, 2023, Groppe and Hundrieser, 2024]. In the present embedded fixed-span model, support-complexity inheritance is exact and the dual smoothness threshold reduces to $\alpha > k/2$. Calibration supplies the only remaining ingredient: evidence that the triangular and i.i.d. rate estimates stay close across the regularization band. That evidence motivates the broader conjecture, but it does not close the bandwise embedded-model statement as a theorem. Additional one-dimensional smooth fixed-support diagnostics show that triangular and i.i.d. fitted exponents can agree closely on some fixed-ε bands, but the agreement is not uniform across the tested roughness levels. The unresolved operational step beyond proposition 2.11 is uniform control across a nontrivial ε-band and higher-dimensional embedded classes. Direct Jacobian diagnostics concentrate that difficulty on moderate compact bands bounded away from zero, where the cross-coupling block stays stable and the self-coupling derivative blocks dominate the instability.

Regular-family inheritance. The broader extension question is whether regular dominated families inherit the same horizon logic with exponents determined by local geometry. The relevant setting is a family $\{P_\theta : \theta \in \Theta\}$ with locally quadratic testing geometry and metric comparability

$$d(P_\theta, P_{\theta'}) \asymp \|\theta - \theta'\|^\alpha$$

for some metric exponent $\alpha > 0$.

Conjecture 2.14 (Regular-family horizon inheritance)**.** *Let d be a probability geometry and let $\{P_\theta : \theta \in \Theta\}$ be a locally regular dominated family for which the i.i.d. benchmark satisfies a metric finite-sample law*

$$\mathbb{E}\, d\Big(\widehat{P}^{(n)}_{\mathrm{iid}}, P_\theta\Big) \le C_K n^{-a}$$

uniformly on the parameter region of interest. Suppose the path obeys a deterministic Hölder envelope in parameter space,

$$\|\theta_{t-j} - \theta_t\| \le \zeta j^H.$$

Then the induced metric staleness should scale as

$$d(P_{\theta_{t-j}}, P_{\theta_t}) \lesssim \zeta^\alpha j^{\alpha H},$$

so the corresponding tracking problem should satisfy the horizon-family member

$$n^*(a, \alpha H) \asymp (C_K/\zeta^\alpha)^{1/(a+\alpha H)},$$

with optimized risk of order

$$C_K^{\alpha H/(a+\alpha H)} \zeta^{\alpha a/(a+\alpha H)}.$$

In the locally regular parametric subcase, the natural metric finite-sample exponent is $a = \alpha/2$, giving the horizon scale

$$n^*(\alpha/2, \alpha H) \asymp (C_K/\zeta^\alpha)^{1/(\alpha(1/2+H))}.$$

Regular dominated families are the natural extension class. Support-changing and singular families remain genuine obstruction classes, so transport-Hölder control alone is too weak to support a single blanket extension theorem.

Together, these settings show how the abstract law extends beyond the one-dimensional proof model while keeping the formal claims within their proper scope.

Once a finite-sample regime and a roughness level are specified, the law yields an explicit memory scale and an optimized error scale. Rougher drift and weaker finite-sample rates both force shorter memory and larger residual error.

3 Empirical Signatures

The empirical section makes the theory visible in finite samples. The aim is to exhibit the main observable signatures of the temporal-validity horizon and to report evidence relevant to the fixed-ε Sinkhorn conjecture.

All experiments in this section are synthetic and fully specified by the stated data-generating designs. The Gaussian U-curve uses 20 Monte Carlo seeds, the roughness-family sweep uses 12, the fixed-ε operational map fits log–log slopes over sample sizes $(24, 48, 96, 160)$ on the pairs $(8, 1)$, $(8, 2)$, $(12, 1)$, and $(12, 2)$ using 24 seeds, and the validity-detection experiment uses a piecewise-ramped drift stream with 12 seeds, a fixed operating window, and both ADWIN and Page–Hinkley detectors. Reported curves are seed-averaged means, while the operational map summarizes fitted finite-sample slopes and marks a cell useful when both exponents exceed the usefulness threshold and their gap stays below the calibration tolerance. The empirical goal is qualitative and structural: persistence of shape, ordering, and sign across repeated runs, not precise constant estimation for every curve.

Four signatures matter. First, finite-memory risk should exhibit an interior optimum. Second, that optimum should move with temporal roughness. Third, the fixed-ε Sinkhorn conjecture should receive structured empirical support rather than a single undifferentiated claim. Fourth, temporal validity can fail before changepoint evidence becomes statistically visible.

3.1 A Finite-Memory Optimum

The Gaussian horizon sweep isolates the variance–staleness trade-off directly. Small windows are variance dominated, large windows are stale, and the minimum lies at an interior horizon.

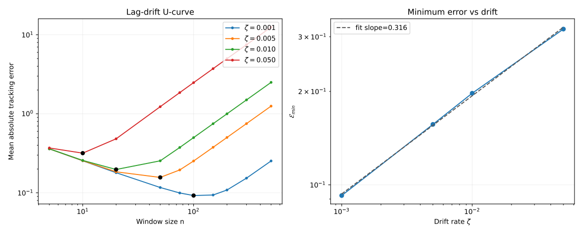

Figure 3: Finite-memory U-curve. Tracking error first falls with horizon and then rises once staleness overtakes the remaining variance gain. The optimum remains finite across drift strengths.

This is the basic empirical signature of temporal validity under drift. Under stationarity, more memory would continue to help. Under drift, the same horizon that reduces variance also ages the evidence.

3.2 Roughness-Dependent Horizon Scaling

The next experiment varies the roughness index directly. For each Hölder-type regime, the surrogate error obeys the same structural decomposition as the theory,

$$\mathcal{E}(n) \approx \sigma n^{-1/2} + \zeta n^H,$$

and the experiment records the empirically optimal horizon across a sweep of ζ values.

The empirical optima separate by roughness class, matching the path-dependent horizon logic of corollary 2.2.

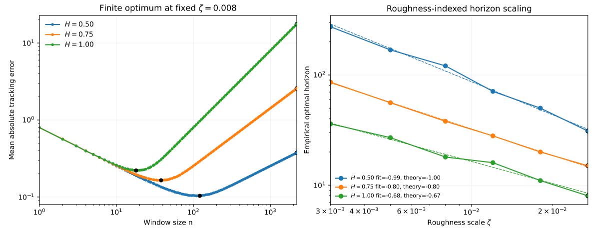

Figure 4: Roughness-dependent horizon scaling. Each path class exhibits a finite optimum, and the location of that optimum changes with H. The horizon is a function of drift regularity, not a fixed window rule.

3.3 Evidence for the Sinkhorn Conjecture

The fixed-ε Sinkhorn experiments address the bandwise question: when a practical geometry replaces raw W_2, does the finite-sample term remain stable enough for the conjectured horizon law to stay informative? The pointwise and bandwise one-dimensional fixed-support statements appear in propositions 2.11 and 2.12; the calibrated embedded fixed-span model shows the regime map for the broader conjectural setting.

The figure below concerns fixed-ε Sinkhorn geometry on embedded low-intrinsic-dimensional support. It shows that the corresponding finite-sample term remains stable across the tested regularization values and that the triangular slope stays close to the i.i.d. benchmark rather than exhibiting a separate exponent. Combined with the structural ingredients isolated in proposition 2.10, this provides empirical support for the bandwise conjecture in conjecture 2.13. In the calibrated fixed-span model with $\alpha = 2$ and span 0.25, the maximal observed stable bands are $\varepsilon_{\max} = 0.5$ for $(d, k) = (8, 1)$, $(8, 2)$, and $(12, 2)$, and $\varepsilon_{\max} = 0.2$ for $(12, 1)$.

The same synthetic family also separates two mechanisms. Raw W_2 comparison between the triangular and i.i.d. empirical measures slows once intrinsic dimension increases, so the embedded fixed-ε statement does not follow from the available W_2 transfer route. Direct Jacobian probes of the centered Sinkhorn scaling map instead isolate a moderate-band regime: on $\varepsilon \in [0.2, 0.8]$, the cross-coupling block stays contractive, while the self-coupling blocks are the main source of instability but appear numerically to plateau with sample size rather than deteriorate. This is evidence about mechanism, not a theorem.

This map gives the shape of the empirical support behind the bandwise conjecture. The observed operational region is neither empty nor uniform across settings; calibration details and supporting tables are kept in appendix B. The tested grid exhibits a stable operational core, with the strongest support on moderate bands bounded away from zero, and the bandwise inheritance law remains conjectural outside the calibrated fixed-span settings shown here.

To probe whether the frontier is controlled more by local intrinsic geometry than by the ambient coordinate count, we estimate the effective intrinsic dimension with TwoNN on the same synthetic frontier family. On the nine-pair, six-ε grid studied here, the resulting $\hat{k}$ predicts the stable-band cutoff better than ambient dimension and

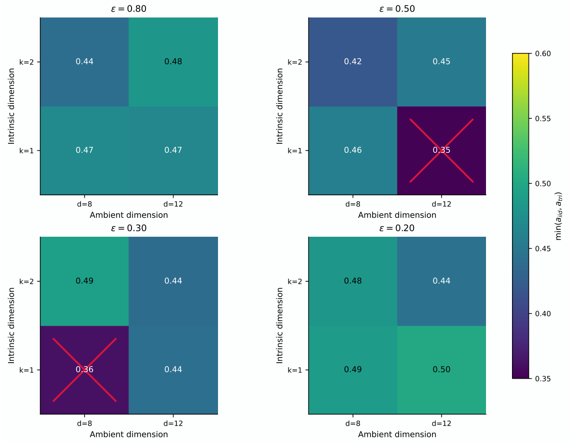

Figure 5: Empirical shape relevant to the bandwise fixed-ε Sinkhorn conjecture. Each panel reports the minimum of the i.i.d. and triangular finite-sample exponents over the ambient/intrinsic pairs tested at one regularization level. Crossed cells mark settings that fail the usefulness criterion. The resulting picture has a stable core and a small mixed boundary region on the tested grid.

improves the binary cut at $\varepsilon = 0.2$. This is consistent with geometry-aware calibration: the relevant control variable is the estimated local geometry, not the ambient coordinate count. The rowwise stability pattern remains mixed, so TwoNN enters here as a regime-selection diagnostic rather than as a replacement for the theorem-level band analysis.

3.4 Validity Loss Before Detector Alarm

Temporal validity and changepoint evidence are different clocks. To make that separation explicit, consider a stream whose local drift rate ramps upward over time. The horizon n_t^* contracts as the ramp steepens, while a long operating horizon remains fixed. Define

$$\Delta_{\mathrm{val-det}} := t_{\mathrm{detect}} - t_{\mathrm{valid}},$$

where t_{valid} is the first time the operating horizon persistently exceeds the temporal-validity horizon, and t_{detect} is the first detector alarm that follows.

Across the reported runs, the measured lag remains positive. Retained evidence can already be invalid for the present while detector statistics remain below alarm threshold.

The positive lag is not a knife-edge feature of one tuning choice. Across the tested ADWIN thresholds, mean detection time remains strictly later than mean validity-expiry time, and the same qualitative lag persists under a different detector family as well. The empirical claim is robustness of the sign of $\Delta_{\mathrm{val-det}}$ on the tested detector/tuning family, not a detector-uniform delay law.

3.5 What the Evidence Shows

Taken together, the experiments support four claims that match the theory. Finite-memory tracking exhibits a structural U-curve rather than an ever-growing memory benefit. The temporal-validity horizon is roughness dependent. The fixed-ε Sinkhorn conjecture receives

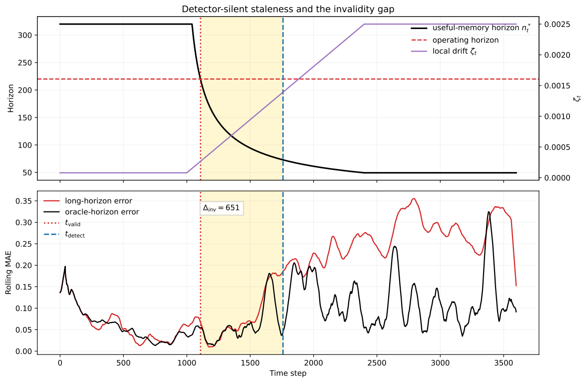

Figure 6: Validity loss before detector alarm. As drift intensifies, the temporal-validity horizon contracts until the operating horizon becomes stale at t_{valid}; the detector responds only later at t_{detect}, leaving a strictly positive validity-detection lag. The lower panel shows excess tracking error accumulating before the alarm appears.

structured empirical support on the tested grid, with the strongest evidence concentrated on moderate bands bounded away from zero. Temporal validity can fail before changepoint evidence appears, producing a measurable validity-detection lag.

4 Related Work

Nonstationary optimization and dynamic regret. Dynamic-regret and variation-budget results show that under drift, window size or forgetting scale must depend on how quickly the target moves [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023, Wang et al., 2024]. Those works are close in spirit, but their primary object is regret under a moving comparator. Here the primary object is different: the temporal validity of retained evidence in a distribution-tracking problem. The horizon is not a tuning parameter attached to regret analysis; it is the statistical object being characterized.

Adaptive windowing and drift detection. Adaptive-window and concept-drift methods ask when accumulated evidence is sufficient to react [Bifet and Gavaldà, 2007, Gama et al., 2004, 2014]. Hinder et al. [2023] emphasize that the relevant window scale is itself part of the problem. The question studied here is adjacent but distinct. Temporal validity is not changepoint evidence: memory can stop representing the present before the available data support a detector alarm. In that sense, the object here is the validity horizon of retained evidence rather than the stopping time of a reaction rule. The validity-detection lag makes that distinction visible in finite samples.

Adaptive filtering and state-space tracking. Classical adaptive filtering and state-space tracking also balance smoothing against responsiveness by tuning forgetting factors, gain schedules, or process noise models [Kalman, 1960, Jazwinski, 1970]. Those analyses usually work under stochastic state-space assumptions and ask for Bayes-optimal tracking of a latent process. The analysis here instead studies deterministic Wasserstein drift envelopes and characterizes a worst-case temporal-validity horizon.

Locally stationary smoothing. Locally stationary time-series analysis likewise studies how window or bandwidth choice must adapt to time variation [Dahlhaus, 1997]. That literature is close in its bias–variance logic, but the moving object there is typically a local spectrum or regression function. Here the moving object is a path of distributions, and the age penalty is measured directly in transport geometry.

Distribution tracking under transport geometry. Wasserstein distance and Sinkhorn-type estimators are natural when drift lives on paths of distributions rather than on scalar losses [Fournier and Guillin, 2015, Boissard and Le Gouic, 2014, Weed and Bach, 2019]. For raw empirical Wasserstein distances, Fournier and Guillin [2015] give the standard ambient-dimension upper bounds, Boissard and Le Gouic [2014] give an earlier covering-based low-complexity bridge, and Weed and Bach [2019] identify sharp finite-sample behavior in terms of Wasserstein dimension. These results are the closest finite-sample precursors to the present law: they show that the exponent in the finite-sample term is geometry dependent even before drift enters.

Entropic and smoothed transport as operational geometries. Regularization changes the statistical geometry as well as the computational one. Entropic and other smoothed transport quantities introduce explicit bias–variance–regularization trade-offs and can recover more stable statistical behavior than raw high-dimensional Wasserstein distances [Genevay et al., 2019, Mena and Niles-Weed, 2019, del Barrio et al., 2023, Rigollet and Stromme, 2025, Stromme, 2023, Groppe and Hundrieser, 2024, Eckstein and Nutz, 2022, Ghosal et al., 2022]. That perspective is exactly why fixed-ε Sinkhorn is treated here as an operational geometry rather than as a surrogate for the raw W_2 theorem. Regularization can induce a usable finite-sample regime on which a finite-memory horizon law still makes sense. In the i.i.d. setting, this literature provides sample-complexity and limit theory for fixed marginals, including fixed-ε differentiability and asymptotic CLTs for the Sinkhorn divergence [Goldfeld et al., 2024], intrinsic- or lower-complexity adaptation, and quantitative stability of regularized transport and Sinkhorn iteration. These fixed-ε asymptotics are the natural i.i.d. anchor for the operational conjecture. Recent Gaussian entropic-OT analyses also make the regularized geometry more explicit in linear-Gaussian settings [Akyildiz et al., 2024]. The Kalman/Riccati structure in that setting suggests a narrower intermediate theorem for streaming Sinkhorn under drift: an explicitly recursive, constructive operational regime between the i.i.d. fixed-ε results and the fully general triangular-array horizon law. A drifted triangular-array theorem that combines sampling, staleness, and optimization control in one horizon law is not yet available. Recent geometry-aware calibration work also points toward intrinsic-dimension-based selection of regularization scales, which is relevant to identifying operational ε-bands in practice [Genans and Wintenberger, 2026]. A concrete minimal-neighborhood route to the same calibration problem is local intrinsic-dimension estimation. TwoNN provides a natural proxy for effective local support geometry and, in the synthetic frontier studied here, a regime-selection diagnostic for operational bands [Facco et al., 2017].

Projection-based and low-dimensional proxy geometries. Sliced, max-sliced, subspace-robust, and projection-robust Wasserstein constructions replace the ambient high-dimensional comparison geometry by one-dimensional or low-dimensional proxies [Kolouri et al., 2018, Deshpande et al., 2019, Paty and Cuturi, 2019, Lin et al., 2020, Nadjahi et al., 2020, Nietert et al., 2022]. These works show that changing the comparison geometry or estimator can materially improve finite-sample behavior for the proxy metric itself. They are therefore relevant here as examples of alternative finite-sample geometries with different exponents, not as automatic improvements for raw W_2.

Wasserstein nonstationarity and robust optimization. Jiang, Li, and Zhang study online stochastic optimization under a Wasserstein nonstationarity measure and obtain distribution-aware regret guarantees [Jiang et al., 2020]. Keehan, Anderson, and Wiesemann analyze related drift-aware weighting questions in Wasserstein distributionally robust optimization [Keehan et al., 2025]. Amini and Vinas give a recent moment-method route to triangular-array concentration in W_1 [Amini and Vinas, 2026]. Those results are adjacent to the finite-sample-regime issue studied here, but they do not by themselves close the W_2 window-mixture concentration problem studied here. The focus is narrower and more structural: for finite-memory tracking, when does retained evidence stop helping because temporal age has become a dominant source of error?

Freshness and Age of Information. The Age-of-Information literature treats age as a resource constraint and asks how update policies balance communication cost against stale information [Kaul et al., 2012, Yates et al., 2021]. This horizon law gives a distributional analogue. Age is costly because it induces geometric error under drift, and temporal roughness determines how quickly that cost accumulates.

Across these neighboring literatures, the closest theme is that nonstationarity makes retention scale matter. The distinguishing claim here is more specific: finite memory is governed by a temporal-validity horizon that balances finite-sample accuracy against drift-induced staleness, and temporal validity can fail before change becomes statistically detectable. The resulting viewpoint is complementary to regret and detection analyses: it characterizes when memory itself ceases to be a statistically valid proxy for the present.

5 Limitations and Scope

The central object is the temporal-validity horizon for finite-memory tracking under drift. The theorem-level results are the general horizon law, the uniform-window staleness bound, the one-dimensional root-n proof model, the structural Gaussian lower bound at the exponent level, and the Gaussian location minimax benchmark. The remaining claims are narrower.

Finite-sample scope. The theory is modular in the finite-sample exponent a, but the main rigorous instantiation is the root-n regime. The tractable proof model establishes that regime in one dimension under bounded support and fixed span. Outside that setting, the same balancing logic continues to hold, but the finite-sample term must come from a different theorem. In particular, no claim is made here for a general raw-W_2 triangular-array theorem in arbitrary dimension.

Operational scope. For fixed-ε Sinkhorn, the structural ingredients of the embedded fixed-span model are proved, and the full triangular-array inheritance law is stated as a conjecture. Exact support-complexity inheritance and the dual threshold $\alpha > k/2$ are proved; the unresolved statement is the full theorem connecting the triangular-array finite-sample term to the i.i.d. benchmark. The numerical evidence is strongest on moderate compact bands bounded away from zero, where direct Jacobian probes remain stable; that evidence does not close the theorem.

Family-level scope. No blanket extension is claimed here to every deterministic W_2-Hölder path class. Continuous regular families and operational geometries can support meaningful finite-sample exponents, but support-changing and singular families behave differently. The natural extension target is therefore a theorem for regular dominated families or operational geometries with comparable local testing structure.

Lower-bound scope. The structural lower bound is subclass based. The Gaussian construction is enough to show that, in the root-n regime, the horizon is structural rather than an artifact of a particular estimator. Class-tightness and constant-sharpness are still unresolved for broader deterministic Hölder path classes and for finite-sample regimes beyond $a = 1/2$. The refined Gaussian geometry sharpens the constants available within that lower-bound program, but it does not close the full distributional lower theory.

Weights and memory shape. Uniform windows are the main case. Nothing here investigates whether broader non-uniform weights can improve constants or alter exponents across richer path classes.

Online roughness estimation. The law is stated in terms of the roughness parameters H and ζ, but no fully online procedure is proposed here for estimating them or for quantifying sensitivity to misspecification. Turning the horizon law into a robust adaptive policy remains open.

Empirical scope. The experiments are synthetic and structurally motivated. Their role is to make the theory visible and to provide evidence for the operational conjecture, not to demonstrate competitive performance on a benchmark suite. Real-data validation and broader comparisons with adaptive horizon-selection rules remain outside the present scope.

Detectability versus validity. Validity loss before detector alarm is an empirical consequence of the theory, not the primary theorem object. Temporal validity can fail before detector alarms appear, but it does not prove a universal detection-delay theorem across detector classes.

Open problems. Each open question below asks for a theorem or construction that would enlarge the theory in a scientifically meaningful direction. The one-dimensional compact-band fixed-ε problem does not appear because proposition 2.12 already settles it; the open operational issue is the embedded bandwise theorem.

Problem 1: sharp finite-sample constants in the proof model. The one-dimensional proof model identifies the correct exponent, but not the best first-moment constant C_K. The open question is to derive a sharp or at least interpretable constant-level bound for the triangular-array finite-sample term, rather than only an $O(n^{-1/2})$ rate. Solving this would sharpen the predicted memory scale inside the first fully proved regime.

Problem 2: lower theory beyond the root-n regime. The structural lower bound proved above is matched to the root-n finite-sample regime. The open question is to identify least-favorable constructions for other finite-sample exponents a, especially in geometries where empirical transport has slower dimension-dependent behavior.

Problem 3: theorem-level regular-family inheritance. A regular-family inheritance principle is stated, but not yet proved it for a broad dominated class. The open problem is to characterize conditions under which local testing geometry and metric geometry together imply a horizon law with finite-sample exponent $a = \alpha/2$ and staleness exponent αH. This would turn the extension principle above into a genuine family- level theorem.

Problem 4: class-tight distributional lower bounds. The Gaussian construction shows that the horizon is structural, but it does not yet deliver class-tight lower bounds for broad deterministic path classes. The open problem is to determine whether the Gaussian benchmark already captures the correct lower rate for larger regular classes, or whether broader constructions are needed. This question is central to understanding how universal the exponent law really is.

Problem 5: a full Gaussian-scale lower theorem. Gaussian scale is a natural benchmark because it changes geometry without leaving the regular parametric world. The open problem is to prove a theorem-level lower bound, and ideally a matching benchmark theorem, for Gaussian scale drift paths. This would clarify how much of the horizon theory depends specifically on location geometry and how much survives under other regular parametric deformations.

Problem 6: bandwise Sinkhorn inheritance. The one-dimensional fixed-ε theory on compact support yields a uniform band law on compact ε-intervals. The open problem is the embedded-model theorem: uniform control of the debiased Sinkhorn rate across a nontrivial $\varepsilon \in \mathcal{B}$ band, with drift, sampling, and optimization error all controlled simultaneously. The numerical evidence is strongest on moderate bands such as $\mathcal{B} \subset [0.2, 0.8]$, where the main unresolved step is uniform control of the self-coupling derivative blocks rather than the cross-coupling block.

Problem 7: optimal weighting beyond uniform windows. The analysis is built around uniform windows, with only brief discussion of other weighting schemes. The open question is whether non-uniform weights can do more than change constants: can they improve the horizon law itself on some path classes, or is the exponent stable under broad classes of memory shapes? A good answer would connect the horizon theory to practical forgetting schemes without collapsing back into heuristic tuning.

Problem 8: online roughness identification and adaptive memory. The law depends on the roughness parameters H and ζ, but no online procedure is given here that estimates them and tracks near-optimal memory in real time. The open problem is to design an adaptive scheme with a provable excess-risk guarantee relative to the oracle horizon. This is the most direct route from the horizon law to an online methodology.

Problem 9: validity and detectability as coupled but distinct objects. The experiments show a positive validity-detection lag, but there is no theorem yet relating temporal validity loss to detector delay. The open problem is to prove detector-specific lower or upper laws that quantify when validity must fail before statistical detection. This would connect the central object to a second, practically important timescale without conflating the two.

6 Conclusion

Under drift, memory is never purely beneficial. It reduces finite-sample noise, but it also ages. The resulting trade-off makes finite memory a temporal-validity object rather than an unlimited statistical resource.

That idea is formalized through the horizon law

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_K n^{-a} + C_S \zeta n^{H},$$

whose optimized scale is

$$n^*(a,H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

The scientific claim is that finite-memory tracking under drift is governed by this balance between finite-sample accuracy and drift-induced staleness.

A tractable one-dimensional proof model supplies a rigorous root-n finite-sample regime, and a structural Gaussian lower-bound construction shows that the corresponding horizon is not an artifact of one estimator. The same object yields a Gaussian location minimax benchmark on deterministic Hölder paths. For fixed-ε Sinkhorn on an embedded fixed-span model, the structural ingredients are proved and the full inheritance law is stated as a conjecture, with numerical evidence concentrated on moderate regularization bands bounded away from zero.

The empirical section makes that object visible through four signatures: a finite-memory U-curve, roughness-dependent horizon scaling, structured evidence for the Sinkhorn conjecture, and a strictly positive lag between validity loss and detector alarm. The resulting message is both theoretical and operational: retained evidence has a path-dependent horizon, and statistical detectability need not coincide with temporal validity.

The main open questions concern sharper finite-sample constants in the proof model, lower theory beyond the root-n regime, broader family-level lower bounds, and theorem-level operational results for the conjectural Sinkhorn extension. In the Sinkhorn direction, the missing ingredient is uniform derivative or equicontinuity control on a moderate compact band.

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A Proof Details

This appendix records detailed derivations for structural results whose full calculations would interrupt the main line.

A.1 Refined Gaussian lower-bound constants

Let

$$S_h := \sum_{r=1}^{h} r^{2H} \qquad \text{and} \qquad G_h := \sum_{r=1}^{h} \big(h^H - (h-r)^H\big)^2.$$

For the Gaussian two-point model with equal priors and common covariance $\sigma^2 I_h$, the Bayes classification error between means $\pm m$ is $\Phi(-\|m\|/\sigma)$. Therefore the ramp and endpoint-minimal payoffs are exactly

$$L_h^{\mathrm{ramp}}(\sigma,\zeta) = \zeta h^H \Phi\Big(-\frac{\zeta}{\sigma}\sqrt{S_h}\Big)$$

and

$$L_h^{\min}(\sigma,\zeta) = \zeta h^H \Phi\Big(-\frac{\zeta}{\sigma}\sqrt{G_h}\Big).$$

Proof of proposition 2.7. Fix $H \in (0,1]$ and write $p_H = 2H/(2H+1)$.

For the ramp profile,

$$S_h \sim \frac{h^{2H+1}}{2H+1}.$$

Set

$$h = A\left(\frac{\sigma}{\zeta}\right)^{2/(2H+1)}$$

with $A > 0$ fixed. Then

$$\frac{\zeta}{\sigma}\sqrt{S_h} \to \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

At the same time,

$$\zeta h^H = A^H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Hence

$$\frac{L_h^{\mathrm{ramp}}(\sigma,\zeta)}{\sigma^{2H/(2H+1)}\zeta^{1/(2H+1)}} \to A^H \Phi\left(-\frac{A^{H+1/2}}{\sqrt{2H+1}}\right).$$

Define

$$x = \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

Then

$$A^H = (2H+1)^{H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$(2H+1)^{H/(2H+1)} x^{p_H} \Phi(-x).$$

Its logarithmic derivative is

$$\frac{p_H}{x} - \frac{\varphi(x)}{\Phi(-x)}.$$

Since the inverse Mills ratio $\varphi(x)/\Phi(-x)$ is strictly increasing on $(0,\infty)$, this objective has a unique maximizer $x_H > 0$, characterized by

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

This proves

$$\sup_{h \ge 1} L_h^{\mathrm{ramp}}(\sigma,\zeta) \sim C_H^{\mathrm{ramp}} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\mathrm{ramp}} = (2H+1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

For the endpoint-minimal profile, the Hölder constraint and endpoint condition imply

$$g_r \ge g_h - (h-r)^H = h^H - (h-r)^H$$

for every feasible endpoint-saturating profile g. Therefore the profile

$$g_r^{\min} = h^H - (h-r)^H$$

minimizes testing energy pointwise and hence minimizes $\sum_{r=1}^{h} g_r^2$. Its rescaled energy satisfies the Riemann-sum limit Thus

$$h^{-(2H+1)} G_h \to I_H,$$

where

$$I_H = \int_0^1 \left(1 - (1-u)^H\right)^2 du = \frac{2H^2}{(H+1)(2H+1)}.$$

With the same scaling

$$h = A \left(\frac{\sigma}{\zeta}\right)^{2/(2H+1)},$$

one has

$$\frac{\zeta}{\sigma} \sqrt{G_h} \to A^{H+1/2} \sqrt{I_H}.$$

Setting $x = A^{H+1/2}\sqrt{I_H}$ gives

$$A^H = I_H^{-H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$I_H^{-H/(2H+1)} x^{p_H} \Phi(-x).$$

The same maximizer x_H therefore yields

$$\sup_{h \geq 1} L_h^{\min}(\sigma, \zeta) \sim C_H^{\min} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

Finally,

$$\frac{C_H^{\min}}{C_H^{\mathrm{ramp}}} = \left(\frac{I_H^{-1}}{2H+1}\right)^{H/(2H+1)} = \left(\frac{H+1}{2H^2}\right)^{H/(2H+1)}.$$

This factor is strictly larger than 1 for $H < 1$ and equal to 1 at $H = 1$. □

A.2 Two-point lower bound used in proposition 2.6

The structural lower bound above is the exponent-level consequence of the exact Gaussian two-point calculation. For the ramp pair

$$\mu_{t-j}^{\pm,h} = \pm\zeta (h-j)_+^H,$$

the present-time separation is $2\zeta h^H$, so any estimator incurs risk at least of order $L_h^{\mathrm{ramp}}(\sigma, \zeta)$. Optimizing over h and using the asymptotic just proved gives the lower scale

$$\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

and therefore the horizon exponent $(2H+1)^{-1}$ for the root-n regime.

B Finite-Sample Diagnostics

This appendix collects finite-sample diagnostics that support the theorem line and the conjectural extensions. These checks are not substitutes for theorem statements.

B.1 Finite-Sample Geometry Check

For the finite-sample term, we estimate empirical W_2 between two i.i.d. samples from the same distribution using exact assignment in low and moderate dimension, then fit a log–log slope in n. We also probe the fixed-span one-dimensional triangular window directly by comparing a triangular-array sample, with one draw from each drifted component, against an i.i.d. sample from the same window mixture. In that check the within-window span ζn^H is held fixed. Finally, we check the Sinkhorn proxy at a few fixed values of ε.

Table 1: Empirical scaling check for the finite-sample term. The i.i.d. low-dimensional W_2 rows stay close to the expected root-n regime, the one-dimensional triangular-array rows compare against the window-mixture benchmark at fixed span, and the Sinkhorn rows illustrate how the proxy changes with ε.

Setting	Estimated slope	Comment
W_2, $d = 1$	-0.48	close to $-1/2$
W_2, $d = 2$	-0.41	slower decay
W_2, $d = 4$	-0.25	slower decay
W_2, $d = 5$	-0.24	slower decay
W_2, i.i.d. mixture, $d = 1$, $H = 0.5$	-0.50	fixed span $\zeta n^H = 0.25$
W_2, triangular, $d = 1$, $H = 0.5$	-0.45	fixed span $\zeta n^H = 0.25$
W_2, i.i.d. mixture, $d = 1$, $H = 1.0$	-0.47	fixed span $\zeta n^H = 0.25$
W_2, triangular, $d = 1$, $H = 1.0$	-0.41	fixed span $\zeta n^H = 0.25$
Sinkhorn $\varepsilon = 0.50$	-0.21	proxy constant changes
Sinkhorn $\varepsilon = 0.10$	-0.53	proxy constant changes
Sinkhorn $\varepsilon = 0.05$	-0.67	proxy constant changes

This table supports the finite-sample interpretation used above. In low dimension, the raw i.i.d. Wasserstein term is compatible with a root-n rate, whereas higher-dimensional raw Wasserstein slows down. In the one-dimensional fixed-span triangular-array check, the recovered slopes stay near the corresponding i.i.d. mixture baseline, which is consistent with the proof-model and design-effect reading used in the 1D proof-model proposition.

B.2 Extended and Operational Regimes

The next table summarizes broader finite-sample evidence beyond the one-dimensional proof model. The extended rows compare triangular and i.i.d. mixture rates on embedded low-intrinsic-dimensional support across span. The operational rows compare fixed-ε Sinkhorn rates across regularization values on embedded low-intrinsic-dimensional support in ambient dimension eight.

Table 2: Representative rows for the extended and operational finite-sample regimes. The extended rows compare triangular and i.i.d. mixture rates across span, while the operational rows compare fixed-ϵ Sinkhorn across ϵ on embedded low-intrinsic-dimensional support.

Regime	Setting	Parameter	Rate exponent a
extended	triangular ambient d=8, intrinsic k=1, span=0.10	0.10	0.4652
extended	iid mixture ambient d=8, intrinsic k=1, span=0.10	0.10	0.5722
extended	triangular ambient d=8, intrinsic k=1, span=0.25	0.25	0.4737
extended	iid mixture ambient d=8, intrinsic k=1, span=0.25	0.25	0.5588
extended	triangular ambient d=8, intrinsic k=1, span=0.50	0.50	0.4792
extended	iid mixture ambient d=8, intrinsic k=1, span=0.50	0.50	0.5425
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.50	0.50	0.5599
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.50	0.50	0.5706
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.20	0.20	0.5466
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.20	0.20	0.6150
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.10	0.10	0.4646
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.10	0.10	0.5060
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.05	0.05	0.4339
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.05	0.05	0.5124

These rows provide calibration for the theorem statements above. The extended-regime rows support the claim that low intrinsic dimension preserves a rate near the root-n regime without a dramatic triangular-versus-benchmark gap. The operational-regime rows supply empirical evidence relevant to the fixed-ε operational setting in definition 2.9: together with exact support-complexity inheritance and the dual threshold $\alpha > k/2$, they support the conjectural inheritance statement discussed in conjecture 2.13. A full proof of triangular-array inheritance would remove this empirical calibration step.

Two complementary diagnostics sharpen the same boundary. First, direct probes of the centered Sinkhorn scaling Jacobian on the embedded fixed-span model show that bands approaching zero are the numerically unstable region, whereas on a moderate band such as $[0.2, 0.8]$ the cross-coupling block remains contractive and the self-coupling inverse norms appear to plateau with sample size. Second, on the supported synthetic frontier grid, TwoNN-based intrinsic-dimension estimates predict the observed stable-band cutoff better than ambient dimension. These diagnostics identify a useful regime proxy and isolate the band where the remaining instability is concentrated. they do not upgrade the embedded fixed-ε statement from conjecture to theorem.

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